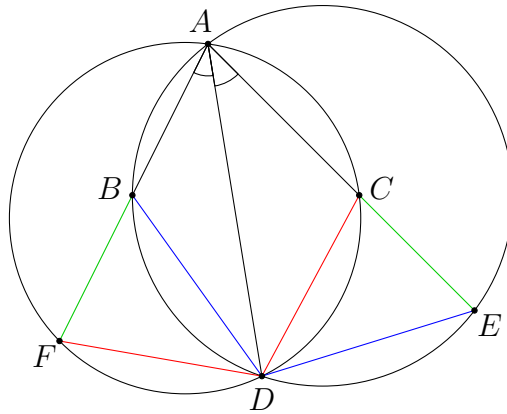


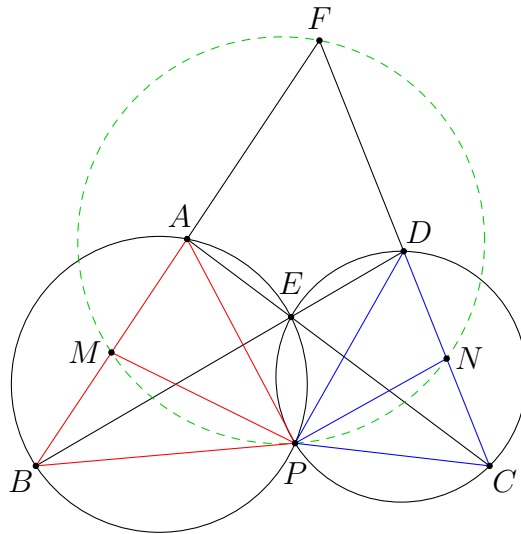
Spiral Similarity — Solutions

Problem 1. In $\triangle ABC$, let D be a point on the internal angle bisector of $\angle BAC$ such that A and D lie on different sides of BC . The circumcircle of $\triangle ABD$ meets the extension of the side AC again at E where C lies between A and E . The circumcircle of $\triangle ACD$ meets the extension of the side AB again at F where B lies between A and F . Prove that $BF = CE$.



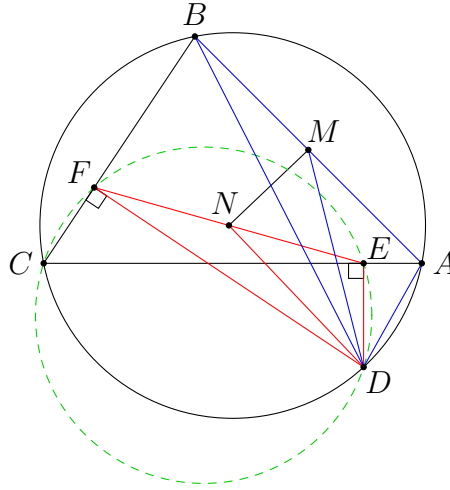
Solution. Note that $\triangle DFB \sim \triangle DCE$ by spiral similarity. Also, as A, F, D, C are concyclic and AD bisects $\angle FAC$, we have $DF = DC$. This implies $\triangle DFB \cong \triangle DCE$, and it follows that $BF = CE$. $\square$

Problem 2. Let $ABCD$ be a convex quadrilateral. The diagonals AC and BD intersect at E . The sidelines AB and CD intersect at F . Let M and N be the midpoints of AB and CD respectively. The circumcircles of $\triangle ABE$ and $\triangle CDE$ meet at E and P . Prove that F, M, P, N are concyclic.



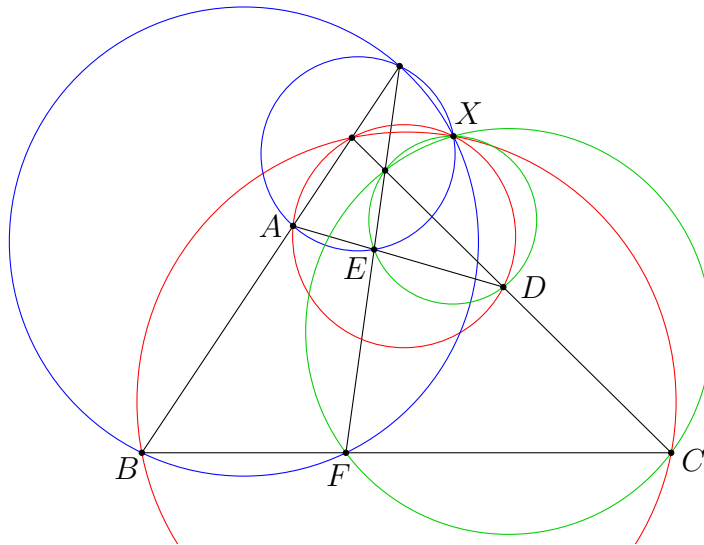
Solution. Note that $\triangle PBA \sim \triangle PDC$ by spiral similarity. Also, M and N are the midpoints of the corresponding sides BA and DC . So P is the centre of spiral similarity sending MA to NC . As MA and NC intersect at F , the points F, M, P, N are concyclic. $\square$

Problem 3. Let $ABCD$ be a cyclic quadrilateral. Let E and F be the feet of projections of D onto AC and BC respectively. Let M and N be the midpoints of AB and EF . Prove that $DN \perp MN$.



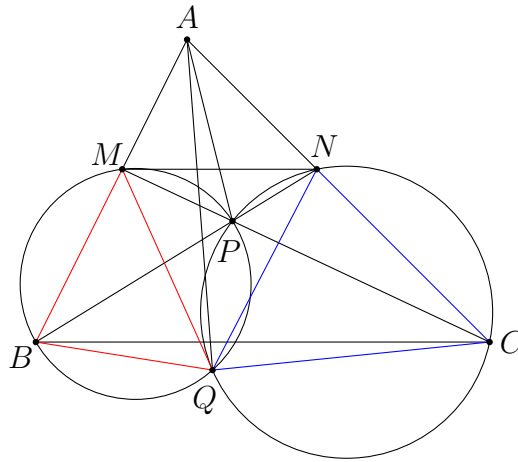
Solution. Since $\angle DEC = \angle DFC = 90^\circ$, the points C, D, E, F are concyclic. As $ABCD$ is also cyclic, we have $\triangle DFE \sim \triangle DBA$ by spiral similarity. Note that N and M are the midpoints of the corresponding sides FE and BA . Therefore, we have $DFNE \sim DBMA$. This implies $\angle DNM = \angle DFB = 90^\circ$. $\square$

Problem 4. Let ABC be a triangle such that AB and CD are not parallel. Let E and F be points on the sides AD and BC respectively such that $AE : ED = BF : FC$. Prove that $ABFE$ and $CDEF$ share the same Miquel point.



Solution. Let X be the Miquel point of $ABCD$. This means X is the centre of spiral similarity sending AD to BC . As $AE : ED = BF : FC$, the points E and F are the corresponding points under this similarity. Thus, X is the centre of spiral similarity sending AE to BF . This shows X is the Miquel point of $ABFE$. Similarly, X is the centre of spiral similarity sending ED to FC , and hence X is the Miquel point of $CDEF$. $\square$

Problem 5. (Balkan MO 2009 Problem 2) Let MN be a line parallel to the side BC of a triangle ABC , with M on the side AB and N on the side AC . The lines BN and CM meet at point P . The circumcircles of triangles BMP and CNP meet at two distinct points P and Q . Prove that $\angle BAQ = \angle CAP$.



Solution. Note that $\triangle QBM \sim \triangle QNC$ by spiral similarity. Therefore, we have

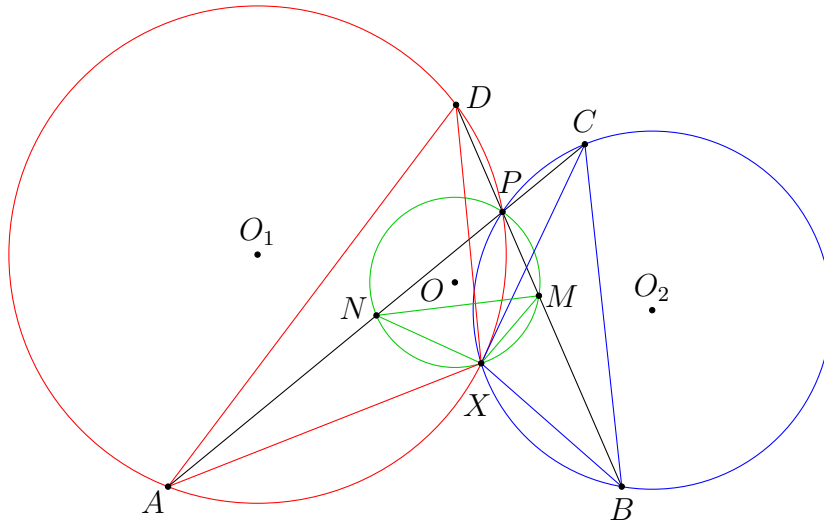
$$\frac{d(Q, AB)}{d(Q, AC)} = \frac{d(Q, BM)}{d(Q, NC)} = \frac{BM}{NC} = \frac{AB}{AC}.$$

The last equality holds since $MN \parallel BC$. This proves AQ is a symmedian of $\triangle ABC$, which is the desired result. $\square$

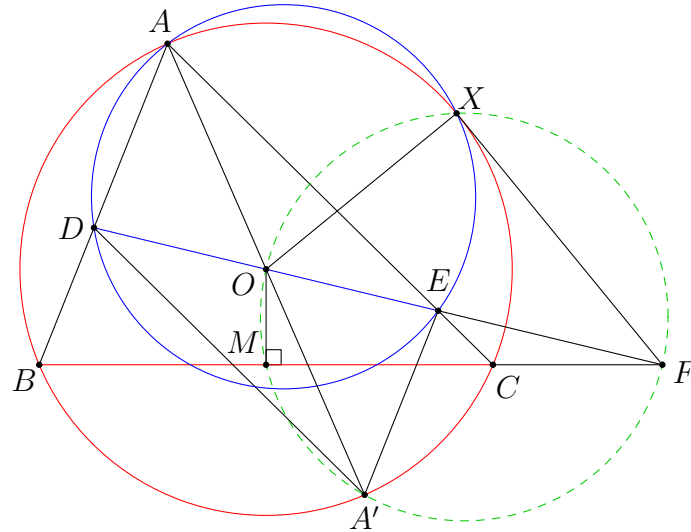
Problem 6. The diagonals AC and BD of a convex quadrilateral meet at P . Let O_1 and O_2 be the circumcentres of $\triangle APD$ and $\triangle BPC$ respectively. Let M, N and O be the midpoints of BD, AC and O_1O_2 respectively. Prove that O is the circumcentre of $\triangle PMN$.

Solution. Let X be the intersection point of (ADP) and (BCP) different from P . Then X is the centre of spiral similarity sending AC to DB . Since N and M are the midpoints of AC and DB , we have $XANC \sim XDMB$. This implies $\triangle XAD \sim \triangle XNM \sim \triangle XCB$. Also, by the spiral similarity, X lies on (PMN) .

Now, let O' be the circumcentre of $\triangle XNM$. Since O_1, O', O_2 are the circumcentres of $\triangle XAD$, $\triangle XNM$ and $\triangle XCB$ respectively, the spiral similarities with centre X send $XADO_1$ to $XNMO'$, and then to $XCBO_2$. This implies $XANC \sim XO_1O'O_2$. Since N is the midpoint of AC , we know that O' is the midpoint of O_1O_2 , which gives $O' = O$ as desired. $\square$

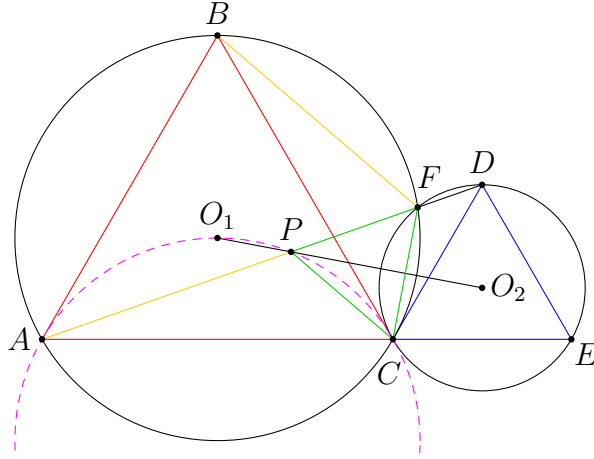


Problem 7. Given $\triangle ABC$, let A' be the point such that $\angle ABA' = \angle ACA' = 90^\circ$. Let D and E be points on the sides AB and AC respectively such that $ADA'E$ is a parallelogram. The lines DE and BC meet at F . The circumcircles of $\triangle ABC$ and $\triangle ADE$ meet again at X . Prove that FX is tangent to the circumcircle of $\triangle ABC$.



Solution. Let O be the circumcentre of $\triangle ABC$, and let M be the midpoint of BC . Note that O is the midpoint of AA' since AA' is a diameter of (ABC) . Thus, O is also the midpoint of DE . Now, we have $\triangle XDE \sim \triangle XBC$ by spiral similarity. This implies $XDOE \sim XBMC$. As OE meets MC at F , the points X, O, M, F are concyclic. Thus, $\angle OXF = 180^\circ - \angle OMF = 90^\circ$. This implies FX is tangent to (ABC) . $\square$

Problem 8. Let C be a point on the segment AE . Two equilateral triangles ABC and CDE are constructed on the same side of AE . Let O_1 and O_2 be the centres of $\triangle ABC$ and $\triangle CDE$ respectively. The circumcircles of $\triangle ABC$ and $\triangle CDE$ meet again at F . The segments O_1O_2 and AD meet at P . Prove that $AP = BF$.



Solution. It is clear that $\triangle CAB \sim \triangle CDE$. So $F = (CAB) \cap (CDE)$ lies on AD (and BE) by spiral similarity. Also, note that $\triangle CAO_1 \sim \triangle CDO_2$. Since AD meets O_1O_2 at P , the points A, C, P, O_1 are concyclic. This implies

$$\angle CPF = 180^\circ - \angle APC = 180^\circ - \angle AO_1C = 180^\circ - 120^\circ = 60^\circ.$$

Also, $\angle CFP = \angle CBA = 60^\circ$. Therefore, $\triangle CPF$ is equilateral. So there is a spiral similarity sending $\triangle CAB$ to $\triangle CFP$, and hence $\triangle CAP \sim \triangle CBF$. Indeed, we have $\triangle CAP \cong \triangle CBF$ since $CA = CB$. This implies $AP = BF$. $\square$

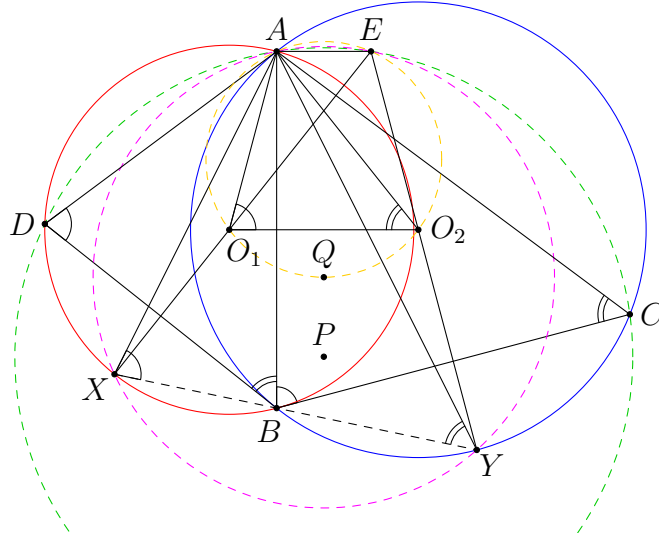
Problem 9. (India MO 2020 Problem 1) Let Γ_1 and Γ_2 be two circles of unequal radii, with centres O_1 and O_2 respectively, intersecting in two distinct points A and B . Assume that the centre of each circle is outside the other circle. The tangent to Γ_1 at B intersects Γ_2 again in C , different from B ; the tangent to Γ_2 at B intersects Γ_1 again at D , different from B . The bisectors of $\angle DAB$ and $\angle CAB$ meet Γ_1 and Γ_2 again in X and Y , respectively. Let P and Q be the circumcentres of triangles ACD and XAY , respectively. Prove that PQ is the perpendicular bisector of the line segment O_1O_2 .

Solution. We only work on the configuration as shown. Firstly, since $\angle ADB = \angle ABC$ and $\angle ABD = \angle ACB$, we have $\triangle ADB \sim \triangle ABC$. The circumcentre O_1 of $\triangle ADB$ corresponds to the circumcentre O_2 of $\triangle ABC$. Note that X and Y are the midpoints of arcs DB and BC respectively. Therefore, we have $ADBO_1X \sim ABCO_2Y$. This implies $\angle AXY = \angle ADB = \angle AXB$. Thus, X, B, Y are collinear.

Let XO_1 meet YO_2 at E . As XO_1E is the perpendicular bisector of BD , and YO_2E is the perpendicular bisector of BC , the point E is the circumcentre of $\triangle BCD$. This implies

$$\angle DEC = 360^\circ - 2\angle DBC = 360^\circ - \angle ADB - \angle DBC - \angle ACB = \angle DAC.$$

So E lies on (ACD) .



Next, since A is the spiral similarity sending O_1O_2 to XY , both AO_1O_2E and $AXYE$ are cyclic quadrilaterals. We claim that AO_1O_2E is an isosceles trapezoid. Since it is cyclic, it suffices to show $\angle O_2O_1E = \angle AO_2O_1$. Indeed, letting $\angle ADB = \angle AO_1O_2 = \alpha$ and $\angle ABD = \angle AO_2O_1 = \beta$, we have

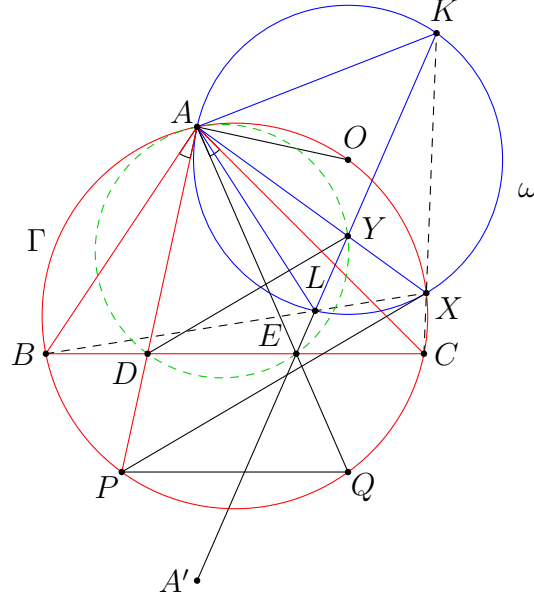
$$\begin{aligned}
 & \angle O_2O_1E = \angle AO_2O_1 \\
 \Leftrightarrow & \angle EO_1A = \alpha - \beta \\
 \Leftrightarrow & 180^\circ - \angle AO_1X = \alpha - \beta \\
 \Leftrightarrow & 180^\circ - 2\angle ABX = \alpha - \beta \\
 \Leftrightarrow & 180^\circ - 2\left(\beta + \frac{1}{2}\angle DAB\right) = \alpha - \beta \\
 \Leftrightarrow & 180^\circ - 2\beta - (180^\circ - \alpha - \beta) = \alpha - \beta.
 \end{aligned}$$

This clearly holds. So our claim is true. Now, since AE is the radical axis of (ACD) and (XAY) , the line joining the centres P and Q of (ACD) and (XAY) is the perpendicular bisector of AE . Thus, PQ is the perpendicular bisector of O_1O_2 . $\square$

Problem 10. (Iran MO 3rd Round 2020 Geometry Problem 2) Triangle ABC with its circumcircle Γ is given. Points D and E are chosen on segment BC such that $\angle BAD = \angle CAE$. The circle ω is tangent to AD at A with its circumcenter lies on Γ . Reflection of A through BC is A' . If the line $A'E$ meets ω at L and K , then prove either BL and CK or BK and CL meet on Γ .

Solution. Let O be the centre of ω . Let X be the intersection point of Γ and ω different from A , and let Y be the intersection point of AX and $A'K$. Also, extend AD and AE to meet Γ again at P and Q respectively. Note that $PQ \parallel BC$ as $BPQC$ is an isosceles trapezoid. We only consider the configuration as shown. We claim that both BL and CK pass through X , which clearly solve the problem.

Note that OP is a diameter of Γ since $\angle OAP = 90^\circ$. Thus, XP is also tangent to ω . It follows (from the limiting case) that the spiral similarity $\mathcal{T}$ with centre A mapping ω to Γ maps the point X to P .



Since $\angle DEA' = \angle DEA = \angle PQA = \angle PXA = \angle DAY$ (the last equality holds since P lies on the perpendicular bisector of AX), the points A, D, E, Y are concyclic. Then we have

$$\angle AYD = \angle AED = \angle AQP = \angle AXP.$$

Thus, $YD \parallel XP$. This implies $AY : YX = AD : DP$, and so $\mathcal{T}$ maps Y to D as well. Also, as $\angle CDA = \angle KYA$ by the concyclic points, $\mathcal{T}$ also maps the line KL to the line BC . Therefore, $\mathcal{T}$ maps K and L to C and B respectively. This proves X lies on both KC and LB by spiral similarity as desired. $\square$